

RF fields, causing interference.

To reduce these problems, a radical approach is required if the bond is to work at high frequencies. The most effective method is to split the bonding process into two—mechanical and electrical. Mechanical part of the bond provides only the physical strength to make it robust mechanically, while electrical part of the bond provides low-impedance connectivity to make the bond electrically robust.

Bonds can be broadly classified as direct bonds and indirect bonds. Direct bonding is where specific portions of the surface areas of the members are placed in direct contact by permanent or semi-permanent bonds. Permanent bonds can be achieved by welding, brazing or soldering. Semi-permanent bonds are bolted connections that provide flexibility and accessibility. For satisfactory bonding, the bolt or screw should serve as a fastener, and should maintain a pressure of  $85\text{kg}/\text{cm}^2$  to  $110\text{kg}/\text{cm}^2$ .

An indirect bond is used in situations where the metal-to-metal contact is not electrically reliable, like for parts that are removed frequently, parts that have relative motion like hinges, parts that use dissimilar metals and parts that are exposed to corrosion. Here, in addition to the primary bond (which provides for the mechanical joint), an indirect bond in the form of straps or jumpers (thick wires) is incorporated (Fig. 22).

Jumpers are employed in low-frequency scenarios and can be short or stranded conductors, generally round in cross-section. These jumpers, unfortunately, exhibit self-inductance and residual capacitance at high frequencies due to the skin effect. In such cases, special flat braids or sheet-metal straps called bond-straps are used. For bond-straps to work effectively, width-to-thickness ratio should be ten or more, while length-to-width

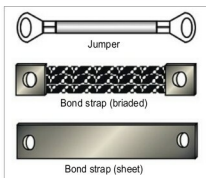


Fig. 22: Indirect bonding devices

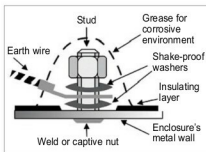


Fig. 23: Bonding of earth wire

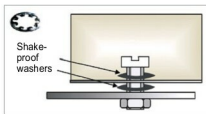


Fig. 24: Bonding of protective wire

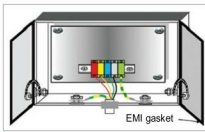


Fig. 25: Bonding in electronic cabinet

ratio should not exceed five.

**Bonding of protective earth wire.** In a typical electronics cabinet, a protective earth wire (green or green-yellow wire) is used for safety reasons to connect the cabinet to external earth.

As shown in Fig. 23, this is usually done by providing a bonding terminal—essentially a bolt screwed to position and held in place by a

captive nut or a stud welded to the enclosure cabinet, the latter being a better option. While fixing the earth wire, any insulating surface finish (paint or powder coating) is removed from the contact area.

The contact area should be slightly larger than the surface area of the lug. A spiky or shake-proof washer is first put into place to ensure a good lifetime bond. Over this is placed the lug. Another shake-proof washer is then placed over the lug after which the nut is tightened. If the bond is likely to be exposed to moisture or corrosive environment, a coating of paint or grease is applied over the bond.

**Bonding of internal protective earth wires.** Earth conductors of modules housed in non-metallic enclosures, installed inside a cabinet, have to be connected to local ground reference or mounting plates or chassis. In such cases, a captive nut-and-screw arrangement is useful as shown in Fig. 24. The captive nut is welded on the other side of the mounting plate, and the earth conductor with a lug is fixed by a screw that slides through this nut.

**Bonding in electronic cabinets.** In any electronic cabinet, direct metal-to-metal contacts/bonds give best RF performance since the contact is over the entire surface rather than a single point. For this reason it is imperative to have local RF reference in the form of a metal plate or sheet, and not a single bolt or stub, as shown in Fig. 25.

The metal plate exhibits low inductance and, hence, provides low impedance at RF frequencies. All modules having a metal enclosure such as filters, switched-mode power supplies and drives are directly installed on this mounting plate, allowing contact over the entire mating surface.

The plate can be connected to either the protective earth coming from the utility or to a dedicated facility earth, the latter being more useful in